
From Inverse Optimization to Sequential Decisions: Conformal Prediction for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) is inverse optimization with a sequential
2 forward problem: demonstrations reveal an unknown reward, and the recovered
3 reward is subsequently optimized to prescribe a policy. This perspective exposes
4 the same weakness identified by Conformal Inverse Optimization (CIO) for one-
5 shot decisions: collapsing heterogeneous objectives to one point estimate can
6 yield a bad downstream decision. We introduce **CP-IRL**, which transfers CIO’s
7 calibrate-then-robustify principle to Markov decision processes. From a popula-
8 tion of demonstrators, CP-IRL conformally calibrates a set of reward parameters
9 consistent with a new demonstrator and optimizes a policy against the resulting
10 reward ambiguity. The sequential setting requires new machinery. We reduce each
11 calibration score to a reward-dimensional convex program using the Bellman resol-
12 vent, and prove that a fixed transition kernel makes the robust MDP convex for any
13 compact reward ambiguity set; a two-state counterexample shows that this property
14 can fail when ambiguity also enters the dynamics. We establish finite-sample
15 coverage, regret bounds, and a separation example in which point-estimate IRL has
16 unbounded regret while CP-IRL’s regret vanishes. Experiments on random MDPs,
17 gridworld, and Objectworld identify the method’s scope: robustification reduces
18 true-reward regret under systematic estimation bias, but incurs an insurance cost
19 when the point estimate is merely noisy and approximately unbiased.

20 1 Introduction

21 Inverse reinforcement learning (IRL) is usually introduced as a distinct learning problem: infer a
22 reward from behavior, then optimize that reward to prescribe a policy. Viewed end to end, however,
23 IRL is a sequential instance of *inverse optimization*. The observed decision is a policy or trajectory,
24 the latent objective is a reward function, and the forward problem is an MDP. This connection is more
25 than terminology. It suggests that uncertainty about the recovered objective should be propagated
26 into the downstream decision, rather than discarded after producing a point estimate.

27 Conformal Inverse Optimization (CIO) formalizes exactly this idea for one-shot problems Lin et al.
28 [2024]. CIO replaces a single recovered objective with a conformally calibrated uncertainty set
29 and optimizes robustly over that set. Its guarantee is finite-sample and distribution-free under
30 exchangeability, and its separation example shows why the change matters: point-estimate inverse
31 optimization can retain unbounded downstream regret even with unlimited data. The natural question
32 motivating this paper is therefore:

Can CIO’s calibrate-then-robustify principle be transferred from one-shot inverse optimization to inverse reinforcement learning without losing coverage or tractability?

We answer this question with CP-IRL. A population of demonstrators supplies the exchangeable calibration units; each observed policy defines a feasible reward set through Bellman optimality; split conformal prediction calibrates a spherical cap around any chosen IRL point estimator; and a robust MDP converts that calibrated reward uncertainty into a deployed policy. Thus, the conceptual pipeline is inherited from CIO, while the technical work lies in adapting and optimizing it for sequential decisions.

Making the CIO template work for IRL is not a mechanical transplant. Two structural obstacles are specific to the sequential setting:

1. **The nonconformity score is no longer an LP-duality fact.** CIO’s per-point calibration score is computable in closed form because the forward problem is a linear program and reward feasibility is characterized by LP dual feasibility. In an MDP, “this policy is optimal for this reward” is a Bellman-optimality condition, not an LP-duality condition, and computing it naively requires jointly optimizing over the reward parameter *and* the policy’s value function – a variable count that grows with the state space.
2. **The robust decision problem is a minimax game over policies, not points.** CIO’s robust forward problem is a single robust linear program. The MDP analogue is a robust MDP: $\max_{\pi} \min_{\theta}$ over an ambiguity set. General robust MDPs with ambiguity that couples across states are NP-hard Wiesemann et al. [2013]; most tractable results in the literature rely on *rectangularity* assumptions that restrict how the adversary’s choice can vary across states.

We resolve both obstacles and develop the MDP counterpart of CIO’s principal guarantees. The contributions are:

- **Transfer: conformal inverse optimization for IRL.** We formulate one demonstrator as one exchangeable inverse-optimization instance, calibrate a reward uncertainty set through Bellman-optimal reward feasibility, and establish finite-sample coverage together with perceived- and true-regret bounds (§3–§5).
- **Optimization: sequential structure without state-sized calibration** (§4.1). For a fixed demonstrator policy, the induced value function is *linear* in the reward parameter via the Bellman resolvent $(I - \gamma P_{\pi})^{-1}$; substituting this collapses the per-demonstrator calibration score from a joint optimization over (θ, V) – size $d + |S|$ – to an optimization over θ alone – size d – with no relaxation, by the policy improvement theorem. We further give a Sherman–Morrison–Woodbury update that recomputes this resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when two demonstrators’ policies differ at only k states, motivated by and tailored to the population-of-demonstrators setting.
- **Tractability: a boundary between reward and dynamics ambiguity** (§4.2). We show the robust MDP is a convex program in occupancy-measure space whenever ambiguity is confined to the reward parameter *for any compact ambiguity set*, via a Bellman-recursion-free argument, and – new relative to the closest existing sufficiency-side result (Gadot et al. Gadot et al. [2024], who solve the convex L_p -ball case via Hölder duality and policy gradient) – we prove this is *tight*: an explicit two-state, two-action MDP with a one-dimensional, convex, non-product ambiguity set coupling reward and transition probability breaks concavity, with a hand-verifiable rational-valued Jensen violation.
- **Scope: robustness helps under bias, not noise alone.** Our sequential analogue of CIO’s separation example proves that classical point-estimate IRL can have diverging regret while the calibrated policy’s regret vanishes. Across random MDPs, a region-based gridworld, and Objectworld, we find that calibrated robust hedging reduces true-reward regret specifically when the point estimate is *systematically* biased (as our degenerate construction manufactures), and *increases* regret when the estimate is merely noisy but unbiased – a precise, falsifiable boundary condition for the method’s benefit that we report as a finding, not a limitation to be hidden.

85 2 Related Work

86 **Inverse optimization and its calibration.** Inverse optimization recovers parameters of an optimization problem from observed decisions (Ahuja & Orlin Ahuja and Orlin [2001]; Chan et al. Chan et al. [2018]). Classical formulations return a point estimate; more recent work constructs data-driven uncertainty sets around it (Chenreddy, Bandi & Delage Chenreddy et al. [2022]; Sun, Liu & Li Sun et al. [2023]), but – as Lin, Delage & Chan Lin et al. [2024] show – none of these give the specific guarantee we need: that the *next* observed decision’s true parameter is covered with a stated confidence, distribution-free. Conformal Inverse Optimization is the first to close this gap using split conformal prediction Vovk et al. [2005], and is the paper this work directly extends from one-shot decisions to MDPs.

95 **Feasible reward sets in IRL.** A separate line of work characterizes the *set* of rewards consistent with observed IRL data (Poiani et al. Poiani et al. [2024]; Lazzati & Metelli Lazzati and Metelli [2025]; Skalse et al. Skalse et al. [2022]) and proves PAC-style bounds on recovering it, using it directly as a robust-MDP ambiguity set. This line shares our target object – a feasible reward set characterized by Bellman-optimality constraints – but its guarantees are PAC bounds requiring a generative model of demonstrator sub-optimality; ours require only exchangeability of demonstrators, no assumption on the *process* by which any individual demonstrator deviates from optimality. The transfer question this line raises – whether a feasible reward set identified on one MDP transfers to another (Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024]) – is a genuinely open problem for our setting that we do not attempt to close here (see §7).

105 **Robust MDPs and the rectangularity boundary.** Robust MDPs with ambiguity coupling across states are NP-hard in general (Wiesemann, Kuhn & Rustem Wiesemann et al. [2013]); tractability is classically recovered via *s*- or *sa*-rectangularity (Iyengar Iyengar [2005]; Nilim & El Ghaoui Nilim and El Ghaoui [2005]), which restricts the adversary to choose independently per state. Recent work extends tractable, *non*-rectangular ambiguity specifically for reward-only uncertainty with a fixed transition kernel: Gadot et al. Gadot et al. [2024] solve the convex L_p -ball case exactly via Hölder duality and a policy-gradient method with proven convergence; Xu et al.’s Group-Robust MDPs Xu et al. [2026] handle *feature-wise* (d-rectangular) ambiguity that couples both reward *and* transition, at the cost of restricting to that rectangular structure. Our sufficiency result (§4.2) covers the same fixed-kernel, reward-only regime as Gadot et al. but for an *arbitrary* compact ambiguity set (not only L_p -balls), via a direct occupancy-measure convex program rather than a policy-gradient method; our necessity result – an explicit demonstration that convexity is lost the moment ambiguity couples into the kernel, even minimally – has no counterpart in either paper.

118 **Conformal prediction in RL-adjacent settings.** Conformal prediction has been used for offline-RL pessimism from Q-value or state-action-coverage uncertainty (UARM, Pan et al. Pan et al. [2026]), for calibrated intervals in infinite-horizon policy evaluation, and for modeling *other agents’ actions* as conformal prediction sets in multi-agent RL (CAMMARL, Gupta & Kahou Gupta and Kahou [2023]) – the last of these is explicitly motivated by an analogy to IRL but never calibrates a reward set. None combine conformal calibration with a reward vector recovered from a population of IRL demonstrators and a robust-MDP regret guarantee, which is the gap this paper closes.

125 **Preference learning and RLHF.** Preference-based and RLHF-style learning fits a reward (or policy) from labeled comparisons or demonstrations, typically from individual expert trajectories (Christiano et al. Christiano et al. [2017]; Rafailov et al. Rafailov et al. [2023]) or crowdsourced preference populations. Our per-demonstrator exchangeable-unit design (§3) is structurally suited to recast “one annotator, one calibration point” for heterogeneous-preference RLHF populations, giving a coverage guarantee of the form “the deployed policy is provably good for at least a γ -fraction of a population whose individual reward functions are never directly observed” – a guarantee we are not aware of in the RLHF calibration literature – though we do not run an RLHF-scale experiment here (see §7).

133 3 Setting

134 We now make the CIO–IRL correspondence precise. We adopt the population-of-demonstrators setting used throughout this work (formalized as Assumption IRL-1 in our coverage proof): a shared tabular MDP $M = (S, A, P, \beta, \phi)$ with known transition kernel P , discount β , and feature map $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing reward $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws

perceived rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim P_\theta$ i.i.d. and each acts optimally in M under their own $\hat{\theta}_k$, producing an observed policy $\pi_k = \pi^*(\hat{\theta}_k)$. This is the direct MDP analogue of CIO’s setting, in which each demonstrator plays the role of one (decision, context) pair – and it is the setting under which CIO’s Theorem 2’s exchangeability argument transplants essentially verbatim (§5). We commit to this “unit A” design over two weaker alternatives – per-task exchangeability with a single demonstrator (still exchangeable, treated as an easy corollary) and per-trajectory exchangeability within one demonstrator (broken by within-trajectory Markov dependence, and used only as an ablation) – and are explicit throughout about what breaks under each relaxation.

The learning task, mirroring CIO’s pipeline, has three steps: (i) split the population into training and calibration sets; (ii) fit a point estimate $\bar{\theta}$ on the training set with any IRL method (we use and compare a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL by MCMC – §6); (iii) calibrate a spherical-cap uncertainty set $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 = 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$ on the calibration set and solve the robust MDP $\max_\pi \min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top \mu(\pi)$ in occupancy-measure space, where $\mu(\pi)$ is π ’s occupancy measure.

4 Method

4.1 Tractable calibration via the Bellman resolvent

CIO’s calibration score for a validation point k is $c_k = \max_\theta \{\theta^\top \bar{\theta} : \theta \in \Theta^{\text{OPT}}(\hat{x}_k, u_k), \|\theta\|_2 \leq 1\}$, where Θ^{OPT} is characterized by LP dual feasibility – tractable because the forward problem is a linear program. In an MDP, the analogous feasible set for a policy π is

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, Q_\theta^\pi(s, a) \leq Q_\pi^\pi(s, \pi(s)) \forall s, \forall a\},$$

the exact set of reward vectors for which π is Bellman-optimal, by the policy improvement theorem. Computing this naively requires the joint optimization $\max_{\theta, V} \theta^\top \bar{\theta}$ subject to the Bellman-optimality inequalities written in terms of both θ and the a priori unknown value function V – a variable count of $d + |S|$.

The key fact we use: for a **fixed** policy π , the policy’s own induced value function V_θ^π is *linear* in θ , via the Bellman resolvent:

$$\begin{aligned} V_\theta^\pi &= (I - \beta P_\pi)^{-1} \Phi_\pi \theta =: W_\pi \theta, \\ Q_\theta^\pi(s, a) &= M_\pi(s, a)^\top \theta, \\ M_\pi(s, a) &:= \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi. \end{aligned}$$

Substituting collapses the calibration score to $c_k = \max_\theta \{\theta^\top \bar{\theta} : \|\theta\|_2 \leq 1, M_\pi(s, a)^\top \theta \leq M_\pi(s, \pi(s))^\top \theta \forall s, a\}$ – an optimization over θ alone (size d), with M_π precomputed once per distinct policy via a single resolvent solve. This is exact, not a relaxation: the reduction is a direct consequence of the policy improvement theorem, verified against a slow joint- (θ, V) reference implementation to solver precision across randomized instances, and against independent value iteration. The amortized speedup over the joint formulation grows with $|S|$ (2–5.5 $\times$ observed at $|S|$ in the low hundreds). When a new demonstrator’s policy differs from an already-resolved one at only k states, the resolvent updates via a Sherman–Morrison–Woodbury identity in $O(k^2|S| + k^3)$ rather than a fresh $O(|S|^3)$ solve.

4.2 A tight tractability boundary for the robust MDP

The robust decision problem, $\max_\pi \min_{\theta \in C} \theta^\top \mu(\pi)$, is a minimax game over policies. We work in occupancy-measure space: $M(P)$ denotes the (fixed, since P is known) polytope of achievable occupancy measures, and the robust objective is $\max_{d \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top d$.

Sufficiency. For *any* nonempty compact ambiguity set $C \subseteq \mathbb{R}^d$ – not only the spherical cap – $g_C(x) := \min_{\theta \in C} \theta^\top x$ is a pointwise infimum of affine functions of x , hence concave, regardless of C ’s shape; composing with the linear map $d \mapsto \Phi^\top d$ preserves concavity; and $M(P)$ is a fixed compact polytope. The robust MDP is therefore a **convex program**, solved without any Bellman recursion or rectangularity assumption – the ambiguity never touches the feasible set $M(P)$, only the objective.

182 **Failure beyond reward-only ambiguity.** The sufficiency argument depends critically on keeping
183 the transition kernel fixed. We exhibit a 2-state, 2-action MDP with a convex, one-dimensional,
184 *non-product* ambiguity set coupling reward $\theta(t) = 1 + t$ and transition probability $q(t) = 1 - t$ for
185 $t \in [0, 1]$. The induced worst-case value, as a function of the agent’s (scalar, single-dimensional)
186 policy parameter $p \in [0, 1]$, is $W(p) = p(1 + t)/(1 - \beta pq(t))$ minimized over t ; a closed-form
187 derivation gives an explicit, hand-verifiable, rational-valued Jensen violation: $W(5/18) = 10/27 \approx$
188 $0.370 < \frac{1}{2}[W(0) + W(5/9)] \approx 0.556$, independently reproduced via numerical optimization rather
189 than only the closed-form derivation. The coupling required to break convexity is minimal – a
190 single scalar t linking reward and transition ambiguity along a line segment, far short of a fully
191 unconstrained non-rectangular adversary.

192 5 Theory

193 We prove five further results, each the direct MDP/IRL analogue of a corresponding CIO theorem,
194 and each independently spot-checked computationally against the implemented solvers rather than
195 trusted as derivation alone.

196 **Coverage.** For the calibrated angle α_γ from Theorem 1’s construction, $\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma$,
197 with the standard split-conformal finite-sample concentration bound $O(\sqrt{\log N_{\text{val}}/N_{\text{val}}})$ on how
198 close the realized coverage is to γ . The proof is a line-for-line transplant of CIO’s own exchangeability
199 argument. A marginal-coverage simulation against the actual calibration code (150 trials per (γ, N_{val})
200 cell) confirms coverage $\geq \gamma$ holds cleanly at every tested combination.

201 **Regret bounds.** We derive the occupancy-measure Lipschitz constant governing how much the
202 value function can change under a bounded reward perturbation, $\nu(d) = \|\Phi^\top d\|_2/(1 - \beta) \leq$
203 $\|\phi\|_{2,\infty}/(1 - \beta)$ (the maximum per-state-action feature norm, verified against 200 randomized MDPs
204 with zero violations), and use it to prove the direct analogue of CIO’s Theorem 3. We are explicit that
205 this is a general worst-case bound, not a tight one: cross-checked against the exact closed-form regret
206 derived for our degenerate construction, the general bound’s looseness grows with the construction’s
207 scaling parameter, from roughly 8–15 $\times$ at the low end of the range tested to roughly 390–430 $\times$ at
208 the high end – a materially worse asymptotic gap than CIO’s own Example 1 exhibits for its own
209 general bound (roughly 2.3–4 $\times$).

210 **Separation.** We construct the direct sequential-decision analogue of CIO’s degenerate Example 1: a
211 single-state, two-action MDP whose reward features are the *negation* of CIO’s original cost features.
212 We prove the point-estimate IRL solution converges to a degenerate direction $\theta_u = (1, u)/\sqrt{1 + u^2}$,
213 not the true θ^* , and that this makes classical IRL’s true-reward regret grow unboundedly as the
214 scaling parameter $u \rightarrow \infty$, while the calibrated robust policy’s regret is bounded and shrinks to
215 zero. This reduced two-action MDP’s robust optimum is a genuine *mixture*, with exact closed form
216 $p^*(u) = u^2/(1 + u^2)$, independent of the calibration angle – derived via Sion’s minimax theorem and
217 independently re-verified against the actual implemented solver to 10^{-6} precision. Figure 1 shows
218 the live 10-seed computational confirmation.

219 **Efficiency.** We confirm, empirically, that the calibrated set is not vacuously wide: with a *unified* noise
220 parameter controlling all sources of demonstrator population variation simultaneously, the calibrated
221 angle shrinks monotonically from 27.7 $^\circ$ to effectively 0 $^\circ$ as demonstrator-population noise vanishes.

222 6 Experiments

223 Our experiments answer three questions: (Q1) does calibration attain its target coverage without
224 becoming vacuous; (Q2) when does robustification improve downstream regret; and (Q3) are the
225 conclusions tied to one IRL estimator or environment? We evaluate three point estimators – a sub-
226 optimality-loss linear program, MaxEnt IRL, and Bayesian IRL (Metropolis–Hastings posterior mean
227 under a Boltzmann-rational demonstrator likelihood) – each combined with three decision rules – the
228 plain point estimate, our calibrated robust policy, and a “concentration baseline” robust policy whose
229 uncertainty-set radius comes from a classical Dvoretzky–Kiefer–Wolfowitz concentration bound
230 rather than the exact split-conformal quantile – across three environments: uncorrelated random
231 tabular MDPs, a region-based gridworld with spatially contiguous (Voronoi) reward features, and
232 Objectworld, the standard IRL benchmark with smooth, continuous distance-to-object features. All

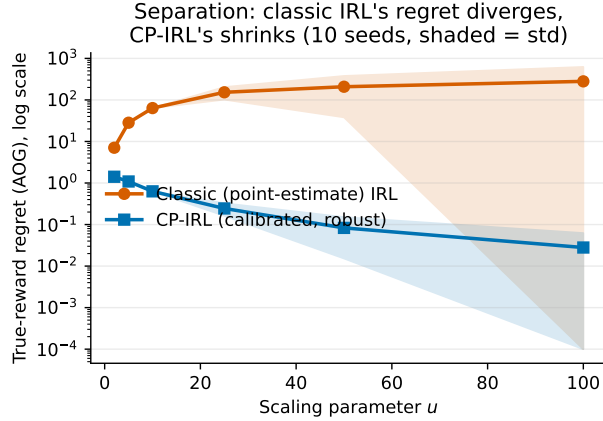


Figure 1: Separation result (§5): true-reward regret vs. the scaling parameter u , log scale, 10 seeds, shaded regions show ± 1 std. Classic IRL’s regret (orange) grows monotonically and without bound; CP-IRL’s regret (blue) shrinks monotonically toward zero over the same range. Variance grows with u for both curves, consistent with tie-breaking sensitivity as $\bar{\theta}$ approaches the exact degenerate boundary.

experiments run on tabular MDPs at a scale ($|S| \leq 36$, $d \leq 5$) chosen to fit within a laptop-class compute budget (20 CPU cores, no GPU dependency, no commercial solver).

Q1: Coverage and efficiency. Realized coverage tracks the target γ across every environment and estimator tested (e.g. ~ 0.88 – 0.89 at $\gamma = 0.8$ on the gridworld), and the calibrated angle responds monotonically and sensibly to γ , to demonstrator noise, and to the train/calibration split ratio. The DKW concentration baseline is consistently and measurably looser than exact conformal calibration at matched nominal confidence, across every estimator and every environment tested.

Q2: The benefit depends on systematic bias. On our degenerate construction (§5), robust hedging dramatically reduces true-reward regret, because the point estimate there is *systematically* miscalibrated by construction. On all three generic environments tested – random MDPs, the gridworld, and Objectworld – under realistic demonstrator-population noise, robust hedging *increases* true-reward regret relative to the plain point estimate, consistently across all three point estimators, holding across a full sweep of the coverage target γ and of the train/calibration split ratio. We verified this is not an implementation defect three independent ways: the robust solver’s reported worst-case value matches an independent second-order-cone-program oracle to solver precision; the robust decision’s worst-case value is confirmably *better* than the plain estimate’s worst-case value over the same calibrated set; and we directly measured that these environments’ optimal policies are sensitive to reward direction – a 14° reward perturbation already changes the optimal action at roughly 10–13% of states across the three environments tested (9.6% Objectworld, 9.9% gridworld, 13.1% random MDPs – a modest, not dramatic, margin). Critically, Objectworld’s lower policy-sensitivity (measured directly, and lower than both the gridworld and random MDPs) did **not** flip the result: robust hedging still costs true-reward regret there, ruling out landscape smoothness as the deciding factor and leaving systematic bias in the point estimate as the consistent, mechanistically understood explanation across all three environments.

Q3: The conclusion persists across estimators and environments. The gridworld comparison combines all three point estimators with the plain, conformal-robust, and DKW-robust decision rules. MaxEnt and Bayesian IRL recover an accurate point direction, whereas the LP estimator has nontrivial error; in all three cases, conformal robustification is less conservative than DKW but incurs higher true-reward regret than the corresponding point policy. The same direction holds in Objectworld. These comparisons separate two claims that are often conflated: conformal calibration can produce a tighter valid uncertainty set than a generic concentration bound, yet optimizing for its worst case need not improve average performance under the true reward.

7 Discussion and Limitations

When does CP-IRL help, precisely? Our theory proves the method dominates classical point-estimate IRL when the point estimate is *systematically* biased. Our experiments show that on generic environments with unbiased-but-noisy point estimates, it does not, and in fact costs regret in exchange for a worst-case guarantee that is real (independently verified) but not free. This is a precise, falsifiable scope condition, not a hedge: the method should be deployed when there is reason to believe point estimation is subject to a genuine structural bias rather than applied by default whenever demonstration data is merely noisy. Characterizing which real-world settings exhibit this kind of systematic bias, and whether it can be diagnosed from data without knowing the ground truth, is an open problem this paper does not resolve.

What is not yet established. The general-case regret bound is provably loose – up to several hundred-fold on the one instance where we can compare it against an exact result – and should be read as a worst-case guarantee, not a predictor of typical-case performance. A source-to-target probe further shows that a reward set calibrated on one MDP does *not* safely transfer to a related MDP with different dynamics: at target coverage 0.8, changing the gridworld slip probability from 0.10 to 0.15 lowers mean target coverage to 0.567 over eight seeds. The probe establishes the failure of naive transfer, not its remedy. Developing a weighted-conformal correction that combines covariate-shift weighting with the principal-angle transferability conditions of Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024] remains future work. We also tested a preference-population/RLHF-style relabeling of our per-demonstrator design: a “response selection” MDP in which the population is reinterpreted as RLHF annotators rather than task demonstrators, using the same calibration and robust-MDP code unchanged. The coverage guarantee transfers exactly as claimed – realized coverage exceeds the target γ at every level tested (0.947–0.994 across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$, 8 seeds) – confirming the framework generalizes by relabeling alone, at the cost of the same limitations documented elsewhere (degenerate low-dimensional calibration geometry in this simple construction; the same systematic-bias-conditional benefit, not a special exemption for the RLHF framing). All experiments are at tabular scale by explicit design choice; we make no claim about function-approximation or deep-RL/language-model settings.

8 Conclusion

IRL is inverse optimization with an MDP as its forward problem. Starting from this connection, we transfer CIO’s calibrate-then-robustify principle to reward learning and supply the sequential machinery the transfer requires: a resolvent-based calibration score, a convex robust-MDP formulation for compact reward ambiguity, and coverage, regret, and separation guarantees. The same connection also clarifies the method’s limits. Calibration certifies uncertainty; it does not make worst-case optimization free. In our experiments, the resulting policy improves true-reward regret when point estimation is systematically biased, but pays an insurance cost when estimation is merely noisy. This distinction is the practical conclusion of CP-IRL and the main condition governing its use.

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